Compositional Inconsistency as a Verification Signal: Pilot Results on MuSiQue

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Building on the verification-budget framework of Allison Li

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Abstract

Compositional inconsistency is the case where a model answers a multi-hop question differently in one step than step by step. It predicts errors on short chains and in weaker models, but as pre-specified it does not add predictive value beyond confidence and self-consistency at scale. We sampled both routes on MuSiQue chain questions with three OpenAI models (gpt-4.1, gpt-4.1-mini, gpt-5.5), labelled every claim against gold answers, and scored selection strategies under the parent project's fixed verification budget (25% of claims checked, swept from 6% to 63%). On 100 two-hop questions, composition was the best non-oracle strategy for all three models, and for gpt-4.1-mini it raised error-prediction AUROC by +0.139 (95% CI +0.035 to +0.250). On 300 questions with 2-, 3- and 4-hop chains, the pre-specified score (maximum over chain regions) showed no gain for either gpt-4.1-mini (+0.001) or gpt-5.5 (−0.007). An exploratory diagnosis traces much of the gpt-4.1-mini null to that aggregator: the full-chain score recovers a +0.053 gain on 3–4 hop chains (CI +0.006 to +0.102). Stated confidence was near chance in every open-book run (AUROC 0.48–0.60). These are pilot results on MuSiQue train; the full-chain hypothesis needs pre-registration and a held-out test.

1 Introduction

The question is whether a structural signal the model cannot misreport can tell a verifier where to spend a limited checking budget better than the model's own confidence. This extends the parent project by Allison Li (think-before-you-trust), which models reasoning as a claim DAG. A check reveals one claim's truth, a caught false claim quarantines everything downstream, and strategies compete under a fixed number of checks. Her simulations use synthetic DAGs with error and propagation rates fitted to PARC (base error 0.176, propagation 0.750). Two of her research questions frame this work:

1. Does self-reported confidence help pick which claims to verify under a limited budget?
2. When confidence is unreliable, is there a structural signal that performs comparably?

The signal tested here comes from category theory. For a chain $x \xrightarrow{f} y \xrightarrow{g} z$, a structure-preserving system satisfies

$$F(g \circ f)(x) = F(g)(F(f)(x)). \tag{1}$$

Answering the composite question directly should therefore agree with answering f, feeding the model's own answer into g, and answering g. The composition score is the total variation distance between the two routes' answer distributions,

$$\delta_{\text{comp}} = d_{\text{TV}}\big(P_{\text{direct}}(a \mid x),\, P_{\text{stepwise}}(a \mid x)\big), \tag{2}$$

and needs no gold answers. The experiment replaces her synthetic DAGs with real MuSiQue chains and real model outputs. It keeps her budget, quarantine rule and reliability metric unchanged, so results are directly comparable to her strategy tables.

2 Methodology

Each question runs through six stages. Only the labelling stage reads gold answers, so every detector signal is gold-free.

2.1 Data

- **Source:** MuSiQue-Ans v1.0 *train*, linear chains only (step k refers to exactly step $k-1$). Dev and test stay untouched for a later held-out study.
- **pilot100:** 100 two-hop chains (seed 0), the same questions as the ongoing Groq gpt-oss-120b pilot.
- **scale300:** 100 each of 2-, 3- and 4-hop chains. The same seed makes its two-hop third identical to pilot100.
- **Condition:** open book by default, so every call sees the same 20 MuSiQue paragraphs (evidence plus distractors). Closed book drops them.
- **Audit:** all questions kept, matching the pilot audit. Pre-registered flags (awkwardly rendered hops, time-sensitive wording) are recorded but not yet reviewed by hand.

2.2 Per-question probes

1. **Original output (1 call):** final answer, an answer per step, and a stated confidence (0–100).
2. **Direct route:** N samples of the composite question.
3. **Stepwise route:** N chains that ask step 1, substitute the model's own answer into step 2, and so on.
4. **Same-route control:** a second independent batch of both routes, so ordinary sampling noise is measured.
5. **Self-consistency:** $3N$ samples of the original question, matched to the probe's call count.

$N = 3$ unless stated, and the temperature is 0.7 for every model. Longer chains get one region per adjacent pair of steps plus the full chain. Sub-regions use a composite built by nesting the step questions, since MuSiQue only writes the full composite.

2.3 Signals

Signal	Definition	Source
Confidence	stated confidence, reversed	self-report
Self-consistency	$1-$ modal share over $3N$ samples	sampling
Same-route noise	mean d_{TV} between the two batches of each route	sampling
Composition (`delta_comp_max`)	max over regions of d_{TV}(direct, stepwise)	structure

Answers are compared after SQuAD-style normalisation (exact match). Abstentions and unparseable replies are kept as explicit outcomes.

2.4 Labels

Every step answer and the final answer are labelled against MuSiQue references. An answer is correct at token F1 ≥ 0.8 (aliases allowed for the final answer) and false otherwise. Abstentions are reported separately and left out of the primary analysis.

2.5 Analyses

- **H1 (prediction):** two logistic regressions with 5-fold out-of-fold predictions. The baseline uses confidence, hop count, question length, self-consistency and same-route noise; the augmented model adds composition. We report the AUROC gain with a 2,000-iteration paired bootstrap 95% CI.
- **H2 (localisation):** AUROC of composition ranking regions that contain a false claim above clean regions.
- **Fixed budget (Stage F):** the parent project's evaluation on real chains. Each step is a node, and a strategy checks the top B claims. A caught false claim quarantines the later steps of its question, and

$$\text{reliability} = 1 - \frac{\text{undetected false claims}}{\text{total claims}}. \tag{3}$$

  The headline B is 25% of claims (her 20 checks of 80 nodes), swept over her 5 to 50 of 80. Strategies are Random (floor, 500 draws), her uncertainty-first and dependency-aware rules ($p_{\text{false}} \cdot D$, with $D = 1+$ downstream claims), self-consistency $\cdot D$, composition $\cdot D$, learned baseline$\cdot D$, learned + composition$\cdot D$, and Oracle (ceiling). Differences carry a 1,000-iteration paired bootstrap over questions.

2.6 Run grid

Task set	Model	Condition	N	Questions
pilot100	gpt-4.1	open	3	100
pilot100	gpt-4.1-mini	open	3, 5, 10	100
pilot100	gpt-4.1-mini	closed	3	100
pilot100	gpt-5.5	open	3, 5	100
pilot100	gpt-5.5	closed	3	100
scale300	gpt-4.1-mini	open	3	300
scale300	gpt-5.5	open	3	300

gpt-5.5 runs with `reasoning_effort: none`, the only setting in which it accepts temperature 0.7.

3 Results

Composition wins the fixed-budget comparison on two-hop pilots, but its pre-specified form adds nothing once 3- and 4-hop chains are included. All numbers come from cached runs replayed with deterministic scoring.

Table 1: Single-signal AUROC and H1 gain (95% CI).

Run	n	Err.	Conf.	SC	Noise	Comp.	H1 gain
pilot100 · gpt-4.1-mini	96	34%	0.48	0.64	0.59	**0.67**	**+0.139** [+0.035, +0.250]
pilot100 · gpt-4.1	100	30%	0.53	0.63	0.66	0.57	−0.026 [−0.064, +0.009]
pilot100 · gpt-5.5	100	34%	0.57	0.63	0.64	0.63	+0.001 [−0.069, +0.064]
scale300 · gpt-4.1-mini	292	40%	0.52	0.67	0.62	0.60	+0.001 [−0.033, +0.036]
scale300 · gpt-5.5	300	32%	0.60	0.63	0.64	0.58	−0.007 [−0.022, +0.008]

3.1 Error prediction (H1)

Table 1 gives each signal's AUROC alone (0.5 is chance) and the AUROC gain from adding composition to the baseline predictor.

Stated confidence is close to chance in every open-book run. The one significant H1 gain, gpt-4.1-mini on two-hop chains, does not survive the move to 300 mixed-length questions.

3.2 Fixed verification budget

Table 2 gives reliability at a budget of 25% of claims. On pilot100, composition beat the parent project's dependency-aware rule by +0.045 (gpt-4.1-mini), +0.049 (gpt-4.1, CI +0.010 to +0.090) and +0.017 (gpt-5.5).

Table 2: Reliability at a 25% verification budget.

Run	Random	Dep.-aware	SC	Comp.	Learned	Learned+C	Oracle
pilot100 · gpt-4.1-mini	0.728	0.755	0.740	**0.790**	0.750	0.770	0.940
pilot100 · gpt-4.1	0.826	0.835	0.865	**0.870**	0.860	0.860	1.000
pilot100 · gpt-5.5	0.825	0.840	0.845	**0.855**	0.820	0.840	1.000
scale300 · gpt-4.1-mini	0.756	0.790	0.753	0.777	**0.796**	0.792	1.000
scale300 · gpt-5.5	0.850	**0.862**	0.860	0.854	0.852	0.857	1.000

At scale the strategies bunch together: the non-random spread is 0.010 for gpt-5.5 and 0.043 for gpt-4.1-mini, where the learned baseline leads. Composition no longer beats dependency-aware (gpt-5.5: −0.015, CI −0.032 to +0.002). Across the full sweep (6% to 63% of claims) no strategy dominates at scale. The leader changes with budget, and the non-oracle strategies stay within 0.045 of each other at every point. Oracle sits below 1.0 for gpt-4.1-mini on pilot100 because it has more false claims than checks.

3.3 Chain length

Composition carries its signal on two-hop chains and fades with length, while confidence improves with length for gpt-5.5 (Table 3).

3.4 Localisation (H2)

Composition ranks regions containing a false claim above clean ones with AUROC 0.64 (gpt-5.5) and 0.59 (gpt-4.1-mini), across 600 regions each. The top 10% of regions are 58% and 73% erroneous respectively, but they cover only 16% and 13% of all erroneous claims.

Table 3: Single-signal AUROC by chain length, scale300.

Model	Hops	Err.	Conf.	SC	Comp. (max)
gpt-4.1-mini	2	34%	0.48	0.64	0.67
gpt-4.1-mini	3	40%	0.52	0.69	0.55
gpt-4.1-mini	4	47%	0.53	0.68	0.52
gpt-5.5	2	34%	0.57	0.63	0.63
gpt-5.5	3	33%	0.62	0.62	0.60
gpt-5.5	4	28%	0.65	0.66	0.56

3.5 Samples per route

More samples do not sharpen the signal. For gpt-4.1-mini on pilot100 the H1 gain is +0.139 at $N = 3$, +0.124 at $N = 5$ and +0.143 at $N = 10$, with composition AUROC flat at 0.67–0.68. gpt-5.5 at $N = 5$ stays null (-0.024).

3.6 Closed book

Without documents, errors rise to 71% (gpt-4.1-mini) and 68% (gpt-5.5). Confidence becomes the best single signal (AUROC 0.73 and 0.65), and composition adds nothing (-0.013 and -0.007). The composition signal is about how a model chains evidence it reads, not about what it knows.

3.7 Inconsistent but correct

At scale, 125 (gpt-4.1-mini) and 126 (gpt-5.5) of 300 questions show route disagreement despite a correct final answer. Only 13 and 20 are wrong with both routes agreeing. Part of this is exact-match answer identity: "May 2, 2008" and "May 2" count as different answers.

3.8 Cost

A two-hop question costs about 28 calls and 66k prompt tokens. The 300-question sets average 61 calls per question (45M prompt tokens per model). Every response is cached, so reanalysis is free.

4 Exploratory finding: the aggregator hides the signal

Much of the gpt-4.1-mini null at scale comes from taking the maximum over regions, not from an absent signal. On 3- and 4-hop chains, sub-regions are asked through synthetic nested composites ("What is the county of (the answer to ...)?"). Those read poorly and come out inconsistent almost every time. For gpt-4.1-mini the mean sub-region score is 0.74 on 3-hop and 0.91 on 4-hop chains, against 0.57 and 0.69 for the full chain, whose direct question is MuSiQue's human-written one. The maximum is then close to saturated and carries little information (AUROC 0.52–0.55 for gpt-4.1-mini on 3–4 hop chains; Figure 1).

Swapping the full-chain score into the H1 model gives Table 4.

This was found after looking at the results, so it is a hypothesis, not a finding. It is worth pre-registering because the mechanism is specific and checkable: the sub-region prompts fail, not the idea of composition. gpt-5.5 stays null under either score.

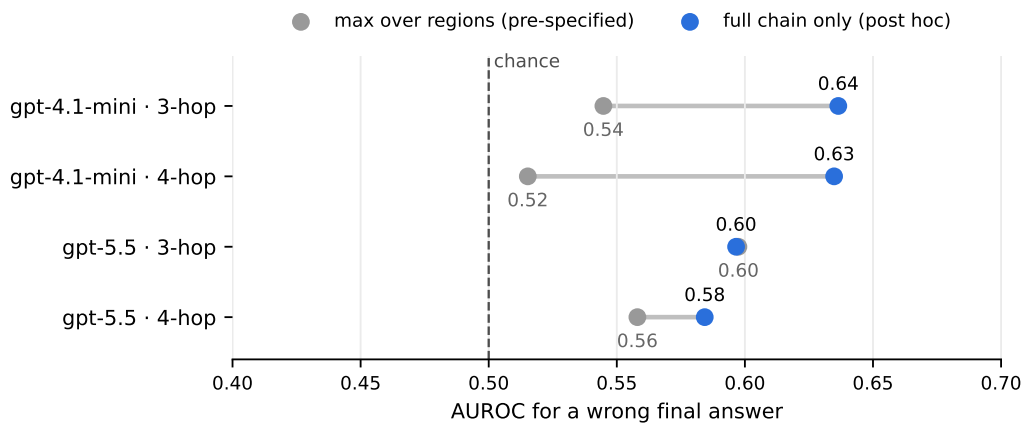


Figure 1: AUROC of the composition score for a wrong final answer on scale300 3- and 4-hop questions (about 100 per row): the pre-specified maximum over regions against the full-chain region alone. Computed after the pre-specified analysis.

Table 4: Exploratory H1 with the full-chain composition score.

Model	Subset	Comp. AUROC (full chain)	H1 gain, 95% CI
gpt-4.1-mini	all 292	0.656	+0.030 [−0.018, +0.077]
gpt-4.1-mini	3–4 hop	0.638	**+0.053** [+0.006, +0.102]
gpt-5.5	all 300	0.598	+0.003 [−0.013, +0.018]
gpt-5.5	3–4 hop	0.587	−0.006 [−0.021, +0.008]

5 Discussion and limitations

The evidence answers the parent project's first question clearly and the second only partly. Stated confidence is a poor guide for choosing checks in every open-book run, so a better signal is needed. Composition is a candidate: it beat confidence-based selection on every two-hop pilot and carries real localisation signal (H2 AUROC 0.59–0.64). But at scale and as pre-specified it is not better than plain self-consistency, which costs the same number of calls and needs no decomposition. Three patterns are consistent across runs:

- **Weaker models show the signal more.** gpt-4.1-mini is where composition adds information. For gpt-4.1 and gpt-5.5, same-route noise and self-consistency already capture it.
- **It is an evidence-chaining signal.** It vanishes in closed book, where confidence takes over.
- **The sample count is not the lever.** $N = 3$ already gives what $N = 10$ gives, which keeps the probe cheap if it is used at all.

Limitations.

- **Answer identity.** Exact-match normalisation counts paraphrases as disagreement. That inflates composition on correct answers (about 125 of 300 questions are inconsistent but correct) and dilutes the signal.
- **Synthetic sub-region prompts.** Nested composites for 3- and 4-hop sub-regions are awk-

ward and saturate the score (Section 5).

- **Labels.** Automatic token-F1 labels miss valid paraphrases and cannot tell an unsupported claim from a valid inference on a false premise. Nothing was hand-labelled.
- **Audit.** 218 of 300 scale300 questions carry audit flags (mostly awkwardly rendered knowledge-base hops), and none were reviewed by hand.
- **One dataset, train split.** Everything is MuSiQue train and no held-out evaluation has been run, so all results are development results.
- **Probe cost sits outside the budget.** Stage F treats probe calls as free alongside a fixed number of checks, while a real deployment pays for both.
- **Temperature.** All models ran at 0.7. gpt-5.5 needed reasoning switched off to allow it, so its reasoning mode is untested.

6 Next steps

Fix the two measurement problems first, then rerun the pre-registered test on held-out data. Only then is the budget-allocation problem worth solving. Steps 1–3 need no new API calls, since every sample is cached.

1. **Answer-equivalence judge (no new probes).** A gold-free judge (alias lists plus a cheap model asked whether two answers name the same thing) merges paraphrases before scoring. Re-score all cached runs and check how many of the roughly 125 "inconsistent but correct" questions remain.
2. **Pre-register the aggregator.** Freeze the full-chain score (or a mean over regions) as the primary composition feature, and use sub-region scores only for H2 localisation. Write it down before step 4.
3. **Hand audit and spot labels.** Review the 218 flagged scale300 questions and hand-label about 50 disagreements, to measure label error and drop broken hop renderings. Re-scoring after the audit is free.
4. **Held-out confirmation.** Draw a fresh set from MuSiQue dev (2-, 3- and 4-hop), run gpt-4.1-mini and gpt-5.5 with the frozen protocol, and test H1 once. Add a second dataset (2WikiMultiHopQA) if the effect holds.
5. **Model sweep.** Add gpt-4o-mini, gpt-5.4-nano, gpt-5.4-mini and the Groq gpt-oss-120b pilot (due around 18–19 November) to test whether the effect tracks model strength.
6. **Budget allocation, only if step 4 is positive.** Put probes and checks in one budget, $B = c \cdot \text{checks} + \text{calls}$, and choose which questions to probe and how many checks remain. The cached $N = 3, 5, 10$ runs let the frontier be computed offline for any cost ratio c. Since $N = 3$ is enough, the useful lever is probe coverage: probe only questions where self-consistency is ambiguous.
7. **Feed real chains back to the parent model.** The measured error and propagation rates on these chains (for example 34–47% final-answer error on gpt-4.1-mini) can replace or check the PARC-fitted constants in `dag_codes/constants.py`.

A Reproducibility

All code is in `will/project1/`. Every API response is cached under `cache/openai/<model>/`, so any analysis below reruns at no cost.

File	What it does
`01_build_pilot.py`	Stage A: select linear chains, render steps, write tasks, gold and audit sheet
`02_generate_original.py`	Stage B: original answer, step answers, confidence
`03_run_probes.py`	Stage C: probes, same-route control, self-consistency
`04_label.py`	Stage D: the only stage that reads gold answers
`05_analyze.py`	Stage E: H1 and H2
`06_budget.py`	Stage F: the parent project's fixed-budget evaluation
`summarize_runs.py`	one table across all runs
`run_openai.sh`	runner: `check`, `replicate`, `build`, `run`, `analyze`, `smoke`
`paper/make_figures.py`	regenerates Figure 1 from the cached runs

```
cd will/project1
./run_openai.sh run pilot100                        # gpt-4.1 by default
TBYT_MODEL=gpt-4.1-mini TBYT_SAMPLES_PER_ROUTE=5 ./run_openai.sh run pilot100
TBYT_MODEL=gpt-5.5 TBYT_REASONING_EFFORT=none ./run_openai.sh run scale300
python3 summarize_runs.py
python3 paper/make_figures.py
```

Each run writes to `outputs_openai/runs/<set>__<model>__<condition>__n<N>/`, with a config snapshot per stage. Total-variation scores are summed in a fixed order and rounded to 10 decimals, so replays are bit-identical. The Groq pilot (`outputs/`) shares the code and is unaffected.